Max COPPIN

PhD

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Work Experience

Now — **Research Fellow,** **Vortex Dynamics group**, *St Andrews*, UK
Oct 2026
- With: Daphné LEMASQUERIER
- Project: Experimental geophysical fluid dynamic: Horizontal centrifugal convection, influence of topography on turbulent zonal jets.

Feb and Jun 2025 — **Academic Visitor,** **AOPP, Oxford University**, *Oxford*, UK
- Supervisors: Peter READ and Bruno DEREMBLE
- Project: Onset of turbulence in horizontal convection experiments

Sep 2026 — **PhD, Institut des Géosciences de l'Environnement (IGE)**, *Grenoble*, France
Oct 2023
- Supervisors: Joël SOMMERIA and Bruno DEREMBLE
- Project: Laboratory observation of rotating free and forced convection

Apr 2023 — **Research internship, Laboratoire d'Océanographie Physique et Spatiale (LOPS)**,
Sept 2023 *Brest*, France
- Supervisors: Xavier CARTON and Camille LIQUE
- Project: Stability of mesoscale vortices under sea ice friction in the Arctic

May 2022 — **Research internship, LOPS**, *Brest*, France
Jun 2022
- Supervisors: Xavier CARTON and Pierre L'HÉGARET
- Project: Front dynamics, between the inflowing Indian Ocean Surface Water, and the outflowing Persian Gulf water, from the observations of 2014

Education

2026–2023 — **PhD in Physical Oceanography**, *LEGI, IGE*, Grenoble

2023–2021 — **Master in Marine Physics**, *University of Western Brittany*, Brest,
M.Sc with highest Honors, rank 2
Specialisation in Geophysical Fluid Dynamics

2021–2020 — **3rd year undergraduate student in Fundamental Mathematics**,
University of Western Brittany, Brest, *B.Sc with Honors, rank 3*
Option in Arithmetic, Combinatorics, and Graph Theory

Publications in peer-reviewed journals

1. Marchand, O., **Coppin, M.**, Negretti, E., Valran, T., Viboud, S., Deremble, B., Sommeria, J. (Submitted.) Laboratory experiments on rotating convection in a stratified fluid at high Rayleigh number *Journal of Fluid Mechanics*

2. **Coppin, M.**, Deremble, B., Sommeria, Negretti, E., J. Valran, T., Viboud, S. (Under Review.) Laboratory observations of a transient Ekman layer *Geophysical and Astrophysical Fluid Dynamics*

3. **Coppin, M.**, Deremble, B., Sommeria, J. (2025) Wind-Mixed Layer Deepening in a Rotating Frame *Geophysical and Astrophysical Fluid Dynamics*, 139. https://doi.org/10.1080/03091929.2025.2609460

Grants - Funding

- Sept 2024 - International mobility grant, **IDEX, Grenoble, France**, €3053

- Apr 2023 - Thematic Schools funding, **ISBlue, Brest, France**, €1300

Scientific Committees

2026 **Scientific Committee Member, GDR MFGA**, Early-Career Researcher Representative

Participation in scientific events

Event organisation

Mar 2026 **Early career GDR MFGA workshop**, *ENS Lyon*, France, Co-organised a one-day workshop (45 participants)

Workshop

May 2026 **Theoretical Challenges in Climate Science**, *Paris*, France, Oral presentation

Jan 2026 **DRAKKAR Ocean Modelling Workshop**, *Grenoble*, France, Poster

June 2025 **International Conference on Geophysical and Astrophysical Fluid Dynamics**, *Plouzané*, France, Oral presentation

Mar 2025 **Parameterizations for global dynamical models in Climatology, Astrophysics and Planetology**, *Les Houches*, France, Oral presentation

May 2024 **Interfaces in the climate system**, *Grenoble*, France, Oral presentation

Nov 2023 **Convection's Days**, *Avignon*, France, Oral presentation

Thematic school

Nov 2025 **Theoretical Challenges for ocean dynamics**, *ENS Lyon*, France

Jan 2024 **Introduction to Data Assimilation**, *Université de Grenoble Alpes*, France

Apr 2023 **Fluid mechanics of planets and stars**, *CISM - Udine*, Italy

Teaching & Supervision

2026 **University of Grenoble Alpes, Grenoble, France**, *Teaching - 3rd year undergraduate*,
Introduction to Physical Oceanography - 12h
Oceanic circulation, Planetary rotation, Ekman Layer, Sverdrup balance

2025 **University of Grenoble Alpes**, *Teaching - 2nd year master student*,
Practical course in Turbulence, diffusion and transport - 5h
Log Layer, Turbulent Entrainment, Ekman Layer

2024 **ENSE3 Engineering school**, *Teaching - 3rd year undergraduate student*,
Practical course in thermal physics - 20h
Forced convection, Conduction, Heat transfer

2024 **LEGI, Grenoble, France**, *Co-supervision (25%) - Lorenzo Blanco*,
Master thesis - 22 weeks
Observation and Parameterization of Oceanic Convection